

Comparing the Mathematical Basis of CellChat and CellPhoneDB for Immune Cell to Cell Communication in Diet-induced Liver Disease

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Abstract

Advances in single-cell RNA sequencing have enabled the inference of cell–cell communication (CCC) from transcriptomic data, leading to the development of numerous computational tools. Here, we compare two widely used CCC inference methods—CellChat and CellPhoneDB—using a shared scRNA-seq dataset from a diet-induced murine model of liver disease. We focus on methodological differences in expression averaging and interaction scoring. CellPhoneDB shows greater sensitivity, detecting more interactions, including those with low gene expression. In contrast, CellChat is more selective, favoring interactions with high and balanced ligand–receptor expression. While CellPhoneDB identifies more significant interactions overall, CellChat’s significant interactions show greater overlap. These findings highlight the impact of algorithmic assumptions on CCC inference and suggest that integrating multiple tools can provide a more comprehensive understanding of CCC in complex disease contexts.

Key words: intercellular communications, single-cell RNA sequencing, Liver Disease, Immunometabolism

Introduction

Cell–cell communication (CCC) is essential to biological processes such as tissue development, immune responses, and repair. While experimental methods like fluorescence imaging, proximity labeling, and gene expression assays offer precise CCC measurements, they are not ideal for large-scale screening [3, 1]. Advances in single-cell RNA sequencing (scRNA-seq) now allow high-throughput, high-resolution analysis of CCC, leading to the development of computational tools designed to infer these interactions from transcriptomic data.

This study focuses on two widely used computational tools for predicting CCC: CellChat and CellPhoneDB. Both follow a similar workflow—filtering low-expression genes, mapping to a curated ligand–receptor (L–R) library, computing average gene expression by cell type, scoring interactions, and performing permutation tests to identify significant interactions [1]. However, they differ in their mathematical implementations of each step. Prior studies have shown low overlap in their predicted interactions when using the same L–R library [4], and benchmarking against validated L–R pairs revealed that while both tools have high specificity, CellPhoneDB exhibits higher precision and sensitivity [16]. A deeper understanding of these methodological differences is essential for selecting the right tool for specific biological questions and for guiding future improvements in CCC prediction tools.

This study has two main goals: (1) To assess the biological reliability of CellChat and CellPhoneDB; (2) To evaluate whether differences in their outputs align with expectations based on their distinct mathematical frameworks.

Using LIANA+ [5], which implements both methods with a shared L–R library, we applied CellChat and CellPhoneDB to single-cell RNA-seq data from liver non-parenchymal cells (NPC) in wild-type and diet-induced Metabolic Dysfunction-Associated Steatotic Liver Disease/Metabolic Dysfunction-Associated Steatohepatitis (MASLD/MASH) mice.

We hypothesized that both tools would detect increased CCC of MHC II pathway in diseased mice and secondly, their differing algorithms would produce distinct patterns. Specifically, we expected: (1) Differences in average gene expression due to their distinct calculation methods; (2) CellChat to favor interactions with balanced, high ligand and receptor expression, and CellPhoneDB to capture those with more imbalanced expression; (3) Variation in the number and overlap of significant interactions reported by each method. Our findings clarify why these tools yield different results and suggest that each has strengths suited to specific CCC research questions.

Methods

Isolation and Single-Cell Sequencing of Liver NPCs

Following euthanasia, mouse livers were perfused *in situ*, mechanically dissociated, and enzymatically digested. The resulting cell suspension underwent low-speed centrifugation (50 $\times$ g, 3 min) to remove debris, followed by enrichment for NPCs. NPCs were filtered through a 40 μ m strainer, and live cells were isolated via fluorescence-activated cell sorting. Only samples with > 90% viability were used for downstream analysis. Sorted NPCs were processed for scRNA-seq using the 10X Genomics Chromium Single-Cell 3' platform, following the manufacturer's protocol.

scRNA-seq Data Processing and Cell Type Annotation

Raw single-cell gene expression count matrices were obtained from 10 mice in 4 groups: SPF mice on a high-fat diet (HF, $n=3$), SPF mice on HF with vancomycin and colistin (HFVHC, $n=3$), germ-free (GF) mice on HF ($n=2$), and GF mice on HFVHC ($n=2$). GF mice lack a gut microbiome; SPF mice have a normal one. Quality control was performed using Scanpy [15], assessing mitochondrial gene content and the number of unique genes per cell. This preprocessing yielded UMI-based gene expression matrices per sample, following standard scRNA-seq guidelines [13]. Highly variable genes (HVGs) were identified using the Seurat v3 method (flavor='seurat.v3') in Scanpy, selecting the top 2,000 per sample [12]. The expression matrix was scaled, and PCA was performed on the HVGs. The first 25 principal components were retained based on the elbow point in the scree plot, where explained variance began to plateau. A k-nearest neighbor graph ($n_neighbors=15$) was constructed in PCA space, with edge weights based on shared nearest neighbors to capture local neighborhood structure. Cell clusters were identified using the Leiden algorithm (resolution = 0.78) [14], yielding 16 clusters. Clusters were visualized using UMAP ($n_neighbors=15$, $min_dist=0.3$) [10], balancing local and global structure. Clusters were annotated using CellTypist [11] with the pre-trained "Healthy_Mouse_Liver.pkl" model. Cell types were assigned by majority vote with a probability threshold of 0.5, resulting in 16 distinct cell types (Fig. 1). These annotations provide biological context for downstream CCC analysis.

Intercellular Communication Inference

To infer CCC, we used the LIANA+ framework [5], which integrates methods including CellChat [9] and CellPhoneDB [6]. For each mouse, normalized log-transformed gene expression matrices were input into LIANA+ and mapped using "mouseconsensus" L-R library. Both methods predicted interactions between all annotated cell types, with significance defined as a permutation test p -value < 0.05.

Comparative and Focused Analyses between CellChat and CellPhoneDB

After running CCC inference for each of the ten mice, we performed several analyses to compare CellChat and CellPhoneDB: **Global Comparison:** We quantified the total number of interactions and significant interactions per mouse identified by each tool. **Analysis of MHC II-T Cell Axis:** Focusing on a known immunological pathway, we extracted scores for the H2-Ab1/H2-Aa $\rightarrow$ Cd4/Lag3

interaction (dendritic cells to CD4⁺ T cells) from both methods and compared them across dietary and gut microbiome groups to assess expected increases in MHC II signaling under disease conditions (HFVHC diet). **Comparison of Average Ligand-Receptor Expression:** For each predicted interaction, we extracted the average ligand and receptor expression levels reported by each method to assess differences in how CellChat and CellPhoneDB average and use expression data in scoring. **Strong Interaction Patterns:** In a representative diseased sample (SPF_HFVHC_137), we computed the absolute expression difference ($|L-R|$) per significant interaction. We then compared the normalized distributions of these differences across methods using total number of interactions CellChat: 24,458; and CellPhoneDB: 36,605. **Shared Interactions:** We compiled and counted all unique significant L-R pairs from $cell_A$ to $cell_B$ across mice and calculated the proportion shared by both tools. These comparisons were visualized using a combined box-and-whisker plot to summarize overlap across experimental groups.

Results

To compare the biological reliability and algorithmic differences between CellChat and CellPhoneDB, we used LIANA+ to analyze L-R interactions from mouse liver NPC scRNA-seq data. Both methods employed the same L-R interaction library ("mouseconsensus") to standardize the comparison. Table 1 summarizes the total number of interactions and the percentage of significant interactions identified for each mouse by both methods. While both CellChat and CellPhoneDB yielded identical total interactions per mouse, CellPhoneDB consistently classified more interactions as significant, indicating it is less selective. Significant interactions identified by CellPhoneDB had scores ranging from 0.043 to 1.80, whereas CellChat's significant interactions scored between 0.00009 and 0.33, reflecting differences in their scoring algorithms.

CellChat and CellPhoneDB are not sufficiently reliable to capture known upregulated pathway in MASLD

During liver pathology progression from inflammation to fibrosis, immune cell crosstalk is crucial in balancing pro- and anti-inflammatory phenotypes. A key upregulated pathway in MASLD is the MHC II pathway, where antigen-presenting cells (APCs) like dendritic cells (DCs) present antigens to CD4+ helper T-cells via MHC II-TCR interactions [20].

We used CellChat and CellPhoneDB to detect MHC II presentation from scRNA-seq data in healthy and diseased states. Interactions of H2-Aa (α) and H2-Ab1 (β) were assessed separately. CD4 and Lag3 served as proxies for TCR expression on regulatory T cells [7, 8].

Overall, neither method showed stronger interactions in diseased mice compared to wild type in either GF or SPF mice (Fig. 2). Lag3 expression was low across all samples, and both methods failed to fully capture MHC II-TCR interactions via Lag3. However, CellPhoneDB consistently detected H2-Aa $\rightarrow$ CD4 and H2-Ab1 $\rightarrow$ CD4 interactions across various dendritic cell subtypes (cDC1s, cDC2s, MigDC2, pDCs), while CellChat identified fewer of these interactions.

When comparing mice that have different gut microbiome backgrounds, CellChat detected fewer DC-CD4+ T-cell interactions in SPF mice compared to GF mice, a trend not seen

with CellPhoneDB. Moreover, CellPhoneDB revealed greater inter-individual variability among SPF mice (Fig. 2).

LIANA’s implementation has discrepancy to CellChat and not CellPhoneDB

During our study, we found that LIANA+’s implementations of CellPhoneDB and CellChat differ in key ways from their original algorithms. We identified these discrepancies by tracing LIANA+ function calls using a Python debugger. For CellPhoneDB, LIANA+ correctly implements all five core steps: (1) gene filtering, (2) averaging expression per cell type, (3) handling multi-subunit complexes, (4) scoring interactions, and (5) assessing significance of interactions (Table 2).

In contrast, CellChat in LIANA+ deviates significantly from the original method [9] (Table 3). First, LIANA+ omits CellChat’s smoothing of expression values using random-walk-based network propagation to project the gene expressions onto high-confidence protein–protein interaction network. Second, It does not consider additional molecular modulators (e.g., agonists, antagonists, cofactors) of L-R interactions. Third, steps (1), (3), and (4) LIANA+’s CellChat differ in implementation: (1): Original CellChat uses a Wilcoxon test to identify differentially expressed genes by cell group ($p < 0.05$), while LIANA+ applies the same filtering as CellPhoneDB. This explains why both tools yield the same number of interactions per mouse (Table 1); (3): CellChat computes geometric means for multi-subunit complexes, but LIANA+’s CellChat uses the same approach as CellPhoneDB; (4): LIANA+ uses a simplified hill function without adjusting for modulatory molecules or sender cells’ and receiver cells’ population sizes.

Due to time constraints, we could not assess further differences in handling heterogeneous L-R complexes. These discrepancies in LIANA+’s implementations caused the outputs from CellChat and CellPhoneDB to deviate from their intended results and limit the scope of meaningful comparisons between the two methods. For clarity, we refer to LIANA+’s versions of these tools as “CellChat” and “CellPhoneDB” throughout the paper.

Given the discrepancies in LIANA+’s implementations, Table 4 summarizes the core differences between CellChat and CellPhoneDB, primarily in how they calculate average gene expression and score interactions. To assess these differences, we first hypothesized the expected outcomes based on each method’s distinct methodology. We then examined variables such as gene expression levels and interaction scores in the output data, assessing whether the results aligned with our hypotheses and any unexpected findings that offer new insights into the two methods. These hypotheses are outlined in Table 3 and are discussed in detail in the following sections.

Average gene expression from both methods retains relative gene expression patterns albeit reporting different levels of magnitudes

CellChat and CellPhoneDB use different approaches to calculate average gene expression for each cell group (Table 4). CellPhoneDB uses a simple arithmetic mean, while CellChat averages the first, second, and third quartiles, making it less sensitive to outliers commonly found in scRNA-seq data [9]. As a result, the two methods can report different expression magnitudes for the same gene. To compare them, we examined statistically significant L-R interactions identified by both methods in a single mouse. We plotted the average expression

of ligands (in sender cells) and receptors (in receiver cells) for each interaction (Figure 3). CellPhoneDB consistently reported higher expression values—approximately three times those from CellChat—likely due to its sensitivity to high outliers. Despite these differences in magnitude, both methods preserved the relative expression patterns between ligand and receptor pairs, indicating that the underlying gene-expression relationships remain consistent.

CellChat Prioritizes Interactions with Minimal Ligand–Receptor Expression Disparity

The second major difference between CellChat and CellPhoneDB lies in how they score L–R interactions (Table 4). CellChat uses a Hill function, commonly applied in systems biology and physical chemistry to model L-R binding process. This function exhibits a threshold-like behavior, sharply increasing scores only when both ligand and receptor expression levels are moderately high. In contrast, CellPhoneDB calculates scores as the arithmetic mean of ligand and receptor expression, producing a linear response without thresholds. As shown in Figure 4 where we plot the scoring functions of the two methods, CellChat assigns near-zero scores when either ligand or receptor expression is low, while CellPhoneDB can yield moderate scores even if one component is minimally expressed. Consequently, we hypothesize that CellChat favors interactions where ligand and receptor expressions are similar and sufficiently high, whereas CellPhoneDB is more permissive of imbalanced expression levels.

We analyzed the distribution of absolute differences in average ligand and receptor expressions for significant interactions identified by CellChat and CellPhoneDB in a single mouse (Fig. 4C). Despite differences in how expression averages are computed that affects the range of absolute differences, CellChat interactions are heavily concentrated near zero expression difference, with frequencies dropping sharply as the absolute difference increases. In contrast, CellPhoneDB yields a broader distribution, including many interactions with large expression disparities. These findings support our hypothesis: CellChat favors interactions with balanced, high expression levels, while CellPhoneDB is more permissive, identifying interactions across a wider expression range.

Comparison of interaction detection: CellPhoneDB captures more significant interactions, and CellChat exhibits higher proportion of overlapped interactions

CellPhoneDB and CellChat both use permutation testing to identify significant L-R interactions (Table 4), randomly shuffling cell type labels 1,000 times and recalculating interaction scores each time to generate a null distribution. A p-value is then calculated as the proportion of permuted scores exceeding the observed (non-permuted) score. We speculated that the differences in scoring functions can generate different synergies between null distributions and tested interaction scores. To explain the logic, we focused on scenarios where the null distribution has low scores, as other combinations are either deterministic or not able to occur. CellChat’s scoring function that has threshold behavior produces null distributions with scores near zero and tested strong and weak interaction scores are more clearly separated, thus altogether making it better to distinguish strong interactions from weak interactions (Fig. 5A). In contrast, CellPhoneDB’s linear scoring leads to null distribution to have higher intermediate scores and leads to

broad, more uniform distribution of tested interaction stores, blurring the distinction between weak and strong interactions (Fig. 5B). These differences likely explain the variation in the number of significant interactions detected and the proportion of overlapping results between the two methods.

The results support expectations: CellPhoneDB consistently reported more significant interactions than CellChat across mice (Table 1) and on average per sender-receiver cell pair (Fig. 5C, Supplementary Data). Across all cell pairs and mice, the overlap in significant interactions accounted for ~55% of CellPhoneDB’s results but ~80% of CellChat’s, consistent with previous findings using different L-R libraries [9]. Due to CellChat’s scoring function, overlapping interactions likely have stronger signals at both ligand and receptor levels. CellPhoneDB’s combination of scoring and null distributions imposes a less stringent cutoff, capturing more weak interactions, aligning with earlier reports of its higher sensitivity [16].

Discussion

Goal 1: Assessing detection of known upregulated pathways in MASLD

Our results indicate that CellPhoneDB is more sensitive than CellChat in detecting low-expression interactions, such as H2-Aa/Ab1 $\rightarrow$ CD4 across dendritic cell types. This may be due to CellPhoneDB’s less stringent filtering, which retains low-signal interactions that CellChat might exclude due to stricter expression or significance thresholds. We did not observe the expected increase in MHC II pathway activity in diseased mice using either method. Additionally, the two methods produced differing levels of variability in L-R interaction scores among mice with the same gut microbiome background, even under identical conditions—highlighting the impact of inter-individual biological noise. Although MHC II upregulation has been reported in MASLD [7], our data showed that variability makes it difficult to conclude whether the pathway is consistently enhanced in the disease condition. We examined how the gut microbiome (GF vs. SPF mice) influences L-R expression in the DC $\rightarrow$ CD4⁺ T-cell axis. CellChat showed reduced detection in SPF mice, a trend not observed with CellPhoneDB, highlighting tool-specific sensitivity differences. Notably, CellPhoneDB results showed greater variability in SPF mice, possibly reflecting microbiome-driven biological heterogeneity.

These findings highlight how methodological differences between CellChat and CellPhoneDB—especially in filtering and scoring—can substantially influence the detection of biologically relevant pathways. Importantly, CCC predictions often assume that ligand and receptor expression levels directly reflect binding strength, overlooking regulatory factors like binding promoters or competitors. Such considerations are critical when interpreting results, particularly in contexts with subtle or variable expression patterns, as seen in MASLD/MASH models.

While we used the well-established MHC II–TCR pathway to assess the biological reliability of CellChat and CellPhoneDB, a key limitation was the lack of a gold standard reference for comparison. Relying on a single pathway also limits the robustness of our evaluation. Future studies should incorporate curated, experimentally validated datasets to better assess method accuracy and biological relevance. Moreover, evaluating

multiple pathway-level cell-cell communication networks can help strengthen the confidence of our conclusions.

Goal 2: Investigate the influence of algorithms in CellChat and CellPhoneDB on yielded output

We used a hypothesis-driven approach to compare CellChat and CellPhoneDB, focusing on how factors like cell type average gene expression, scoring methods, and calculating significance shape their outputs. Our analysis reveals how each tool’s underlying algorithms influence the interactions they identify.

Both CellChat and CellPhoneDB use similar methods to calculate average gene expression, preserving relative patterns despite differences in magnitude—indicating robustness in this step. We hypothesized that differences in their mathematical designs would lead to distinct L–R interaction profiles, and our results support this. CellChat favors interactions with similarly high expression of both ligand and receptor, highlighting strong, balanced signals. In contrast, CellPhoneDB is more permissive, often detecting interactions even when expression is imbalanced, suggesting greater sensitivity. Biologically, such imbalances can still support functional interactions—for example, the NMDA receptor’s ability to bind multiple ligands despite modest expression levels [17, 2]. Thus, CellPhoneDB casts a wider net, while CellChat prioritizes more stringent, potentially higher-confidence interactions.

We found that CellPhoneDB identified more significant interactions than CellChat, likely due to differences in how each method defines and interprets significance based on the gap or synergy between real interaction data and randomized data. Most interactions detected by CellChat were also found by CellPhoneDB, but not vice versa—CellPhoneDB identified many additional L–R pairs. This asymmetry reflects CellPhoneDB’s broader, more inclusive scoring approach, which captures weaker interactions or interactions with imbalanced expressions that CellChat tends to exclude. Rather than contradicting each other, the tools reflect different assumptions: CellChat prioritizes high-confidence signals, while CellPhoneDB favors comprehensive detection. Each has its strengths—CellChat for robust and high-confidence interactions, CellPhoneDB for exploratory or hypothesis-driven analyses of interactions.

Interestingly, most of CellChat’s predicted interactions overlapped with those from CellPhoneDB. A previous study [9] reported that CellChat achieved a higher true positive rate and lower false positive rate compared to CellPhoneDB, suggesting greater accuracy in identifying biologically relevant interactions. Benchmarking studies often assume that more accurate methods will show greater overlap with others [9]. Given CellChat’s high overlap and established accuracy, it is likely the more reliable tool for identifying high-confidence, robust interactions, whereas CellPhoneDB’s broader sensitivity may include more false positives.

Given the complementary strengths of CCC tools, we recommend applying multiple inference methods in parallel to gain a more comprehensive view of CCC. For example, running both CellChat and CellPhoneDB on the same dataset allows users to identify consistently predicted interactions. Aggregation frameworks like LIANA+ can further enhance robustness by integrating results across tools and ranking interactions by consensus [5]. This approach increases confidence in shared signals while still capturing additional candidates from more sensitive methods like CellPhoneDB.

Mouse	Cellchat		CellphoneDB	
	Total # of interac- tions	%p < 0.05	Total # of interac- tions	%p < 0.05
GF_HF_6B	75767	30.6%	75767	47.8%
GF_HF_13L	48687	39.4%	48687	60.4%
SPF_HF_118	86692	33.5%	86692	45.3%
SPF_HF_131	57476	37.3%	57476	55.0%
SPF_HF_133	74449	29.5%	74449	44.6%
GF_HFVHC_5A	85347	27.9%	85347	44.5%
GF_HFVHC_16L	43793	37.9%	43793	57.8%
SPF_HFVHC_124	68322	29.8%	68322	44.5%
SPF_HFVHC_136	90075	30.7%	90075	43.5%
SPF_HFVHC_137	76635	31.9%	76635	47.8%

Table 1. Number of Total and Proportion of Significant Interactions per Mouse Identified by CellChat and CellPhoneDB

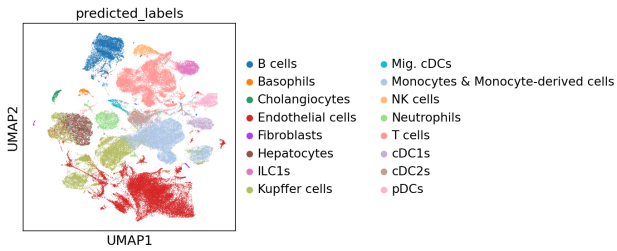


Fig. 1. UMAP of scRNA-seq data of all the 10 mice with each cell colored by a cell type annotated by CellTypist.

Despite the insights gained, our analysis has several limitations. The partial implementations of CellChat and CellPhoneDB within LIANA+ restricted our ability to assess the impact of alternative preprocessing steps, such as gene expression filtering, L-R expression smoothing, and inclusion of modulatory molecules (e.g., agonists, antagonists). We also did not evaluate multimeric ligand-receptor complexes, which are handled differently by each method. Finally, uncertainties in the quality of the mouseconsensus L-R resource used in LIANA+ add further limitations to our findings.

Looking ahead, several directions could extend this work. A key next step is to run the full versions of CellChat, CellPhoneDB, and other CCC tools using a shared L-R library to confirm whether our findings hold under each tool's native settings. Expanding the comparison to include additional methods could reveal whether new trade-offs emerge.

Beyond hypothesis-driven analysis, future studies might explore modular experiments—such as swapping scoring functions or significance criteria between tools—to isolate the impact of specific components on performance. For example, applying CellChat's scoring within CellPhoneDB's framework (or vice versa) could clarify the roles of scoring strategies versus pipeline design. These hybrid tests would help disentangle the source of observed differences and inform the development of improved, integrative CCC tools.

Acknowledgments

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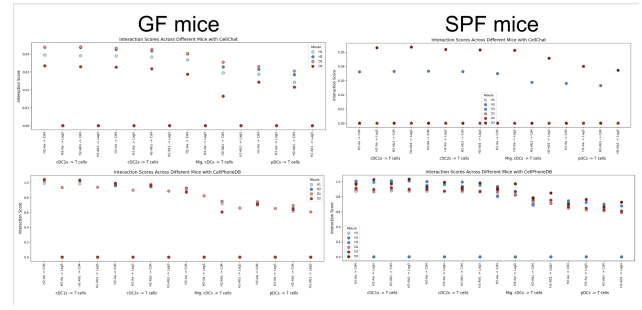


Fig. 2. Comparison of CellChat and CellPhoneDB in detecting MHC II interactions between dendritic cells and CD4⁺ T cells. Interaction scores for H2-Aa/H2-Ab1 (dendritic cells) to CD4/Lag3 (T cells) were extracted per mouse. Mice were grouped into four categories: H1, H2 (GF healthy, N=2); H3 (SPF healthy, N=3); D1, D2 (GF diseased, N=2); and D3 (SPF diseased, N=3).

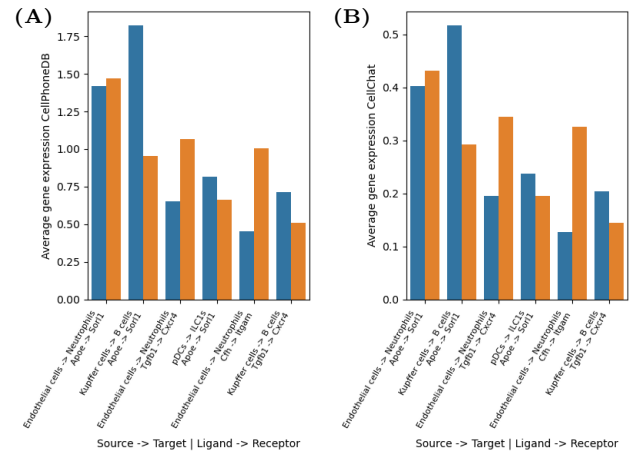


Fig. 3. Average gene expression levels for significant interactions identified by both CellPhoneDB and CellChat in mouse SPF_HFVHC_137. Interactions are labeled as source→target cell | ligand→receptor on the x-axis. Expression levels reported by (A) CellPhoneDB and (B) CellChat.

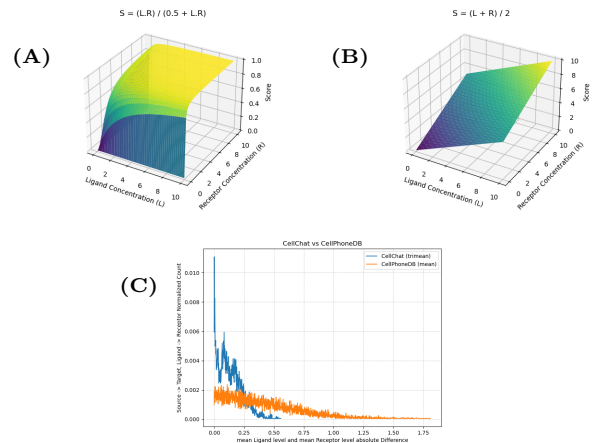


Fig. 4. (A) Interaction scores vs. ligand and receptor expression based on CellChat's scoring function ($K_h = 0.5$). (B) Same plot using CellPhoneDB's scoring function. (C) Normalized distributions of $|L-R|$ average expressions for significant interactions identified by CellChat (blue) and CellPhoneDB (orange) in mouse SPF_HFVHC_137.

#	Main steps of algorithm	CellPhoneDB Paper [6]	LIANA+'s CellPhoneDB
1	Gene filtering	For each potential $L_i \rightarrow R_j$, $Cell_A \rightarrow Cell_B$, requires L_i in $> 10\%$ of cells ($Cell_A$) AND R_j in $> 10\%$ of cells ($Cell_B$).	✓
2	Averaging expression per cell type	$EM = \frac{1}{n} \left(\sum_{i=1}^n x_i \right)$	✓
3	Handling multi-subunit complexes	Use the average expression of the least-expressed subunit.	✓
4	Scoring interactions	$S_{CPDB}(A, B; L - R) = \frac{\bar{E}_A(L) + \bar{E}_B(R)}{2}$	✓
5	Assessing significance of interactions	Permutation test: Randomly permuting cell type labels 1000 times and recalculate $L \rightarrow R$ interaction scores from $Cell_A \rightarrow Cell_B$. $p - value = \frac{\{\#m P_{i,j}^m > P_{i,j}, m = 1, 2, \dots, M\}}{M}$	✓

Table 2. Comparison between CellPhoneDB and its implementation in LIANA+

#	Main steps of algorithm	CellChat Paper [9]	LIANA+'s CellChat
1	Gene filtering	Select differentially expressed genes (Wilcoxon test, $p < 0.05$)	✗ Different method, used CellPhoneDB's method
2	Averaging expression per cell type	$EM = \frac{1}{2}Q_2 + \frac{1}{4}(Q_1 + Q_3)$	✓
3	Smooth out gene expressions	Project gene expressions to a protein-protein interaction database through Network propagation	✗ Absent
4	Handling multi-subunit complexes	$L_i = \sqrt[m]{L_{i,1} \dots L_{i,m1}}, R_j = \sqrt[m]{R_{j,1} \dots R_{j,m2}} \cdot \frac{1 + RA_j}{1 + RI_j}$	✗ The equation is absent. Same as in CellPhoneDB, the average expression of the least-expressed subunit was used.
5	Scoring interactions	$P_{i,j}^k = \frac{L_i R_j}{K_h + L_i R_j} \cdot \left(1 + \frac{AG_i}{K_h + AG_i}\right) \cdot \left(1 + \frac{AG_j}{K_h + AG_j}\right) \\ \times \frac{K_h}{K_h + AN_i} \cdot \frac{K_h}{K_h + AN_j} \times \frac{n_i n_j}{n^2}$	✗ Most of the equation absent except this hill function $P_{i,j}^k = \frac{L_i R_j}{K_h + L_i R_j}$
6	Assessing significance of interactions	Permutation test $p - value = \frac{\{\#m P_{i,j}^m > P_{i,j}, m = 1, 2, \dots, M\}}{M}$	✓

Table 3. Comparison between CellChat and its implementation in LIANA+

#	Main steps of algorithm	CellPhoneDB	CellChat	Expectation/hypothesis
2	Averaging expression per cell type	$EM = \frac{1}{n} \left(\sum_{i=1}^n x_i \right)$	$EM = \frac{1}{2}Q_2 + \frac{1}{4}(Q_1 + Q_3)$	CellChat and CellPhoneDB report different average expression levels for the same gene due to differing sensitivity to outliers in their averaging methods.
4	Scoring interactions	$S_{A,B;L-R} = \frac{\bar{E}_A(L) + \bar{E}_B(R)}{2}$	$P_{ij}^k = \frac{L_i R_j}{K_h + L_i R_j}$	CellChat favors balanced ligand and receptor expression, while CellPhoneDB tolerates larger expression differences.
5	Assessing significance of interactions	Permutation test $p - value = \frac{\{\#m P_{i,j}^m > P_{i,j}, m = 1, 2, \dots, M\}}{M}$		CellChat and CellPhoneDB differ in how test scores relate to their null distributions, leading to varying numbers of significant interactions and differing overlap between methods.

Table 4. Key CellPhoneDB vs. CellChat Steps Driving Output Differences and Resulting Hypotheses.

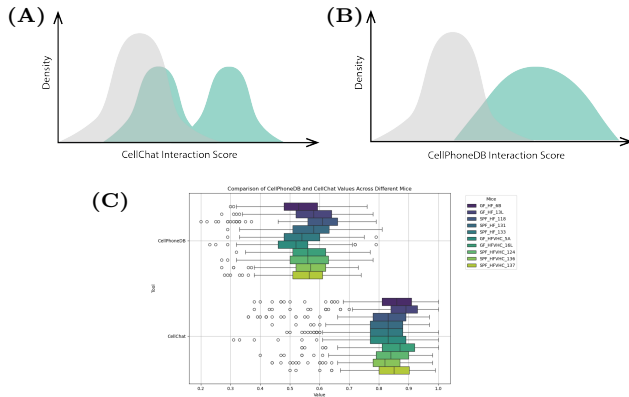


Fig. 5. Illustration of statistical testing dynamics and overlap between CellChat and CellPhoneDB. (A–B) Conceptual schematic of how test interaction scores (cyan) compare to permuted null distributions (grey) in CellChat and CellPhoneDB. Distributions are illustrative and not based on real data. (C) Proportion of significant interactions shared between CellPhoneDB and CellChat for each cellA→cellB pair across mice.

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Data and code availability

Detailed information on software versions and key analysis code are provided in the Supplementary Materials.

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